QUESTION BANK

INDUSTRIAL MANAGEMENT

Module I: Introduction

SAQ (each carry 1 marks):

Sl No	Questions	CO	BTL
1	Which function the basic of all other functions in an organisation?	2	1
2	What is the primary goal of decision making?	2	1
3	Define planning.	2	1
4	Define controlling.	2	1
5	What do you mean by management?	2	2
6	What do you understand by job satisfaction?	2	1
7	What is morale?	2	1
8	Define industrial management.	2	1
9	What is the primary objective of industrial management?	2	1
10	What is span of control?	2	1
11	What is delegation of authority?	2	1
12	What do you mean by organization structure?	2	1
13	Define centralization.	2	1
14	Define decentralization.	2	1

Questions carrying 5 marks:

Sl No	Questions	CO	BTL
1	Write down the functions of management.	2	1
2	Write a short note on span of control.	2	2
3	What is industrial management? Explain the function of industrial management.	2	2
4	State the difference between industrial management and production management.	2	1
5	What do you mean by organization structure? What are the types of organization structures? (3+2)	2	2
6	State the factors affecting industrial relations.	2	2
7	What is job satisfaction and what is the importance of job satisfaction to an organization?	2	2
8	Explain the major factors influencing job satisfaction in an organization.	2	2
9	Discuss the factors affecting organizational culture.	2	2

10	Explain the differences between centralization and decentralization in industrial management with suitable examples.	2	3

Questions carrying 15 marks:

Sl No	Questions	CO	BTL
1	What is organization structure? Briefly explain the types of organization structure with diagram. (5+10)	2	3
2	What do you mean by management. Explain the functions of the management. (5+10)	2	3
3	Define the concept of industrial relation. State the factors affecting industrial relations. (5+10)	2	3
4	Explain the factors affecting employee morale and discuss how to overcome these in organizational context.	2	2
5	Explain the concept of organizational culture and state the difference between centralization and decentralization.	1	1

Module II: PERT/CPM

SAQ (each carry 1 marks):

Sl No	Questions	CO	BTL
1	Define event.	2	1
2	Define activity.	2	1
3	Define dummy activity.	2	1
4	Define network.	2	1
5	What do you mean by CPM?	2	2
6	Define critical activity.	2	1
7	What is PERT?	2	1
8	Define slack or float in PERT.	2	1
9	What is crashing?	2	1
10	What are the applications of crashing?	2	1
11	What is earliest start time?	2	1
12	What is latest finish time?	2	1
13	Define Pessimistic time.	2	1
14	Define optimistic time.	2	1
15	What do you understand by ‘most likely time’?	2	2
16	What is Gantt Chart?	2	1

Questions carrying 5 marks:

Sl No	Questions	CO	BTL
1	What are the key differences in PERT and CPM? What are the advantages of PERT and CPM? (3+2)	2	1
2	How do you calculate the earliest start time and latest finish time for activities in CPM? How do you identify non-critical paths in a PERT or CPM network? (2+3)	2	2
3	Compare and contrast the PERT and CPM techniques in terms of their underlying principles, assumptions, and applications.	2	2
4	Define pessimistic time, optimistic time and most likely time & establish the relationship between them. (3+2)	2	1
5	Explain the concept of crashing with its applications.	2	2
6	Describe the impact of uncertainties and risks on project scheduling in PERT and CPM. Discuss strategies or techniques that can be employed to mitigate these risks and ensure successful project completion. (2+3)	2	2
7	Discuss the concept of resource leveling in context of PERT and CPM.	2	1
8	Can you provide an example of a real-world project where PERT or CPM would be beneficial in scheduling and management?	2	2

Questions carrying 15 marks:

Sl No	Questions	CO	BTL

<table>
<tr>
<td>1</td>
<td>a) Draw the project network. Find out the time, variance and standard deviation of the project with the following time estimates in weeks: (3+3+3+3)
<table>
<tr><th>Activity</th><th>Optimistic time estimate (to)</th><th>Most likely time estimate (tm)</th><th>Pessimistic time estimate (tp)</th></tr>
<tr><td>1-2</td><td>3</td><td>6</td><td>9</td></tr>
<tr><td>1-6</td><td>2</td><td>5</td><td>8</td></tr>
<tr><td>2-3</td><td>6</td><td>12</td><td>18</td></tr>
<tr><td>2-4</td><td>4</td><td>5</td><td>6</td></tr>
<tr><td>3-5</td><td>8</td><td>11</td><td>14</td></tr>
<tr><td>4-5</td><td>3</td><td>7</td><td>11</td></tr>
<tr><td>6-7</td><td>3</td><td>9</td><td>15</td></tr>
<tr><td>5-8</td><td>2</td><td>4</td><td>6</td></tr>
<tr><td>7-8</td><td>8</td><td>16</td><td>18</td></tr>
</table>
b) In a construction project, an engineer estimates that the optimistic time for laying the foundation is 10 days. If the most likely time is 15 days and the pessimistic time is 20 days, what is the expected duration for laying the foundation using PERT? (3)</td>
<td>2</td>
<td>3</td>
</tr>
<tr>
<td>2</td>
<td>A project consists of seven activities and the time estimates of the activities are furnished as under:</td>
<td>2</td>
<td>3</td>
</tr>
</table>

<table>
<tr><td></td><td>
<table>
<tr><th>Activity</th><th>Optimistic Days</th><th>Most likely Days</th><th>Pessimistic Days</th></tr>
<tr><td>1-2</td><td>4</td><td>10</td><td>16</td></tr>
<tr><td>1-3</td><td>3</td><td>6</td><td>9</td></tr>
<tr><td>1-4</td><td>4</td><td>7</td><td>16</td></tr>
<tr><td>2-5</td><td>5</td><td>5</td><td>5</td></tr>
<tr><td>3-5</td><td>8</td><td>11</td><td>32</td></tr>
<tr><td>4-6</td><td>4</td><td>10</td><td>16</td></tr>
<tr><td>5-6</td><td>2</td><td>5</td><td>8</td></tr>
</table>
i. Draw the network diagram.
ii. Identify the critical path and its duration.
iii. What is the probability that the project will be completed in 5 days earlier that the critical path duration?
</td><td></td><td></td></tr>
<tr><td>3</td><td>Draw the network diagram. Determine the earliest and latest times, the total float for each activity, the critical activities, the slacks of the events and the project completion time.
<table>
<tr><th>Name of Activity</th><th>Predecessor Activity</th><th>Duration (Weeks)</th></tr>
<tr><td>A</td><td>-</td><td>8</td></tr>
<tr><td>B</td><td>A</td><td>13</td></tr>
<tr><td>C</td><td>A</td><td>9</td></tr>
<tr><td>D</td><td>A</td><td>12</td></tr>
<tr><td>E</td><td>B</td><td>14</td></tr>
<tr><td>F</td><td>B</td><td>8</td></tr>
<tr><td>G</td><td>D</td><td>7</td></tr>
<tr><td>H</td><td>C, F, G</td><td>12</td></tr>
<tr><td>I</td><td>C, F, G</td><td>9</td></tr>
<tr><td>J</td><td>E, H</td><td>10</td></tr>
<tr><td>K</td><td>I, J</td><td>7</td></tr>
</table>
</td><td>2</td><td>3</td></tr>
<tr><td>4</td><td>Conduct a comparative analysis of PERT and CPM in the context of project management. Discuss the underlying principles, methodologies, advantages, and limitations of each technique. (8+7)</td><td>2</td><td>2</td></tr>
</table>

Module IV: Production Planning and Control (PPC)

SAQ (Each questions carry 1 mark):

Sl No	Questions	CO	BTL
1	What is the primary objective of production planning and control?	4	1

2	Which technique involves breaking down the production process into smaller, more manageable tasks?	4	1
3	Which tool is commonly used for visualizing the flow of work in production processes?	4	1
4	What is the purpose of material requirement planning (MRP)?	4	1
5	Which approach focuses on continuously improving processes and products to achieve customer satisfaction?	4	1
6	What is the primary goal of production control?	4	1
7	What is the purpose of a Bill of Materials (BOM)?	4	1
8	Which production planning tool is used to determine the optimal sequence of tasks in a production process?	4	1
9	What does the term "lead time" refer to in production planning and control?	4	1
10	Which quality management technique focuses on improving processes by identifying and correcting defects?	4	1
11	Which production planning technique uses historical data and statistical methods to predict future demand?	4	1
12	In the context of production planning and control, what does the term "bottleneck" refer to?	4	1
13	Define EOQ.	4	1
14	Define scheduling.	4	1

Questions carrying 5 marks:

Sl No	Questions	CO	BTL
1	Discuss the importance of capacity planning in production planning and control.	4	1
2	What do you understand by the term ‘bottlenecking’? What are the ways to reduce bottlenecking in context of PPC? (2+3)	4	2
3	Explain the concept of Just-In-Time (JIT) production and its significance in modern manufacturing.	4	2
4	Discuss the key principles of JIT and the benefits and challenges associated with its implementation.	4	1
5	Compare and contrast push-based and pull-based production systems.	4	2
6	Briefly explain the function and importance of PPC.	4	2
7	Write down the advantages and disadvantages of push-based and pull-based production systems.	4	2
8	What is Gantt Chart? What is the purpose of Gantt Chart? Write down the limitations of Gantt chart. (1+2+2)	4	2

Questions carrying 15 marks:

Sl No	Questions	CO	BTL

1	Explain the challenges and opportunities associated with implementing Just-In-Time (JIT) production in a manufacturing organization. Discuss the key principles of JIT and the potential benefits of its adoption. (7+8)	4	2
2	Compare and contrast various quality management methodologies such as Total Quality Management (TQM), Six Sigma, and Lean manufacturing. Evaluate their effectiveness in improving product quality, reducing defects, and enhancing customer satisfaction. (9+6)	4	2
3	What do you mean by production planning? Discuss its elements. Discuss the different types of production systems. (2+5+8)	4	2
4	Explain the parameters that must be considered before finalizing the production system. Write down the advantages and disadvantages of JIT. Briefly discuss the objectives and principles of scheduling. (5+5+5)	4	2
5	What are the different types of production system? How does production planning differ from production control? State the functions of production planning. (4+5+6)	4	2

Module VI: Enterprise Resource Planning (ERP)

Questions carrying 1 mark

Sl No	Questions	CO	BTL
1	Define JIT.	6	1
2	Which kind of process is JIT?	6	1
3	What is the basic principle of JIT?	6	1
4	What concept does JIT manufacturing rely on to ensure smooth production flow?	6	1
5	Which metric is often used to measure the effectiveness of JIT manufacturing?	6	2
6	What is the basic objective of JIT?	6	1
7	What does ERP stand for?	6	1
8	What are the key components of an ERP system?	6	1
9	What are the main phases of ERP implementation?	6	1
10	What is the major problem of ERP	6	1
11	What are the needs of logistics?	6	1
12	What is the basic need of ERP for management?	6	1
13	What is the role of top management commitment for implementing ERP?	6	1
14	In which type of production system JIT is required?	6	1

Questions carrying 5 marks

Sl No	Questions	CO	BTL
1	Compare and contrast the fundamental principles of Enterprise Resource Planning (ERP) and Just-In-Time (JIT) manufacturing.	6	1
2	Discuss the risks associated with ERP implementation in JIT manufacturing contexts and strategies to mitigate them.	6	2
3	Describe the importance of ERP in management.	6	2
4	Write down the applications of ERP.	6	1
5	Briefly discuss the benefits of supply chain management.	6	2
6	Briefly explain the components of supply chain management.	6	2
7	What are the advantages of JIT?	6	2
8	Discuss the impact of ERP systems on supply chain management.	6	2

Questions carrying 15 marks:

Sl No	Questions	CO	BTL
1	Compare and contrast traditional supply chain management practices with Just-In-Time (JIT) methodologies. Evaluate the advantages and disadvantages of each approach in terms of inventory management, responsiveness, and cost-effectiveness. (8+7)	6	2
2	Discuss the impact of ERP systems on supply chain management, including inventory control, demand forecasting, and supplier relationship management.	6	2
3	Explain the fundamental principles and key components of supply chain management (SCM) within the context of modern business operations.	6	2
4	Discuss the role of logistics in modern business operations and its significance in achieving competitive advantage.	6	2
5	Discuss the concept of ERP and explain the advantages and disadvantages of ERP. Write down the importance of ERP in management system. (4+6+5)	6	2
6	Write a short note on JIT and explain its advantages. (5+10)	6	2

Module III: Materials Management

Questions carrying 1 mark

Sl No	Questions	CO	BTL
1	1. Materials management mainly focuses on	3	1

	A. management of raw material or components required for continuous production B. production of finished goods and it's sale in the appropriate market C. management of logistics and supply chain activities for timely market reach D. distribution of materials to the seller and distributor for smooth functioning of the market activities		
2	Raw Materials and WIP can be classified under _________ A. Indirect Material B. Direct Material C. Finished Material D. Standard Parts	3	1
3	........................is a buffer stock level or safety stock level under which the stock should not be allowed to fall. A. Average stock level B. Maximum stock level C. Minimum stock level D. none	3	1
4	The optimum level of inventory is popularly referred to as the .... A. Minimum stock level B. Re-order stock level C. Economic Order Quantity D. none	3	1
5	The minimum stock level is calculated as A. Reorder level – (Normal consumption x Normal delivery time) B. Reorder level + (Normal consumption x Normal delivery time) C. (Reorder level + Normal consumption) x Normal delivery time D. (Reorder level + Normal consumption) / Normal delivery time	3	1
6	Define lead time.	3	1
7	Average stock level can be calculated as A. Minimum stock level + 1⁄2 of Re-order level B. Maximum stock level + 1⁄2 of Re-order level C. Minimum stock level + 1/3 of Re-order level D. Maximum stock level + 1/3 of Re-order level	3	1
8	What do you understand by EOQ?	3	1
9	Which of the following is not an inventory? A. Machines B. Raw material C. Finished products D. Consumable tools	3	1
10	What is re-order level?	3	1

11	What is safety stock?	3	1
12	What is inventory?	3	1

Questions carrying 5 marks:

<table>
<tr><th>Sl No</th><th>Questions</th><th>CO</th><th>BTL</th></tr>
<tr><td>1</td><td>Discuss the purpose of Material Control.</td><td>3</td><td>2</td></tr>
<tr><td>2</td><td>Explain the concept and importance of EOQ?</td><td>3</td><td>2</td></tr>
<tr><td>3</td><td>What is the importance of material management?</td><td>3</td><td>1</td></tr>
<tr><td>4</td><td>What are the types of inventory?</td><td>3</td><td>1</td></tr>
<tr><td>5</td><td>Write a short note on ABC analysis.</td><td>3</td><td>1</td></tr>
<tr><td>6</td><td>Write a short note on VED analysis.</td><td>3</td><td>1</td></tr>
<tr><td>7</td><td>Discuss level of inventory control system.</td><td>3</td><td>1</td></tr>
<tr><td>8</td><td>P. Ltd. uses three types of materials A, B & C for production of X, the final product.
The relevant monthly data for the components are as given below
<table>
<tr><th></th><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Normal usage</td><td>200 units</td><td>150 units</td><td>180 units</td></tr>
<tr><td>Minimum usage</td><td>100 "</td><td>100 "</td><td>90 "</td></tr>
<tr><td>Maximum usage</td><td>300 "</td><td>250 "</td><td>270 "</td></tr>
<tr><td>Reorder Quantity</td><td>750 "</td><td>900 "</td><td>720 "</td></tr>
<tr><td>Reorder period (in month)</td><td>2 to 3</td><td>3 to 4</td><td>2 to 3</td></tr>
</table>
Calculate for each component:
(a) Re-order level
(b) Minimum level
(c) Maximum level
(d) Average stock level</td><td>3</td><td>1</td></tr>
<tr><td>9</td><td>The following is a summary of the recepits and issues of materials in a factory during the month of April.
<table>
<tr><th>Date</th><th>Particulars</th><th>Qty.</th><th>Rate per unit Rs.</th></tr>
<tr><td>1</td><td>Received</td><td>2000</td><td>10</td></tr>
<tr><td>5</td><td>Received</td><td>300</td><td>12</td></tr>
<tr><td>8</td><td>Issued</td><td>1200</td><td>-</td></tr>
<tr><td>10</td><td>Received</td><td>200</td><td>14</td></tr>
<tr><td>12</td><td>Issued</td><td>1000</td><td>-</td></tr>
<tr><td>23</td><td>Received</td><td>300</td><td>11</td></tr>
<tr><td>31</td><td>Issued</td><td>200</td><td>-</td></tr>
</table>
Prepare a statement showing the pricing of issues on the basis of -
a) FIFO method
b) LIFO method
c) Simple Average method
d) Wighted Average method</td><td>3</td><td>1</td></tr>
</table>

Module 5: Value Analysis (VA) and Cost Control

Questions carrying 1 mark

Sl No	Questions	CO	BTL
1	Define Value Analysis.	5	1

2	Define value.	5	1
3	What is function cost matrix?	5	1
4	What do you understand by value index?	5	1
5	Value analysis examines the a) Design of every component b) Method of manufacturing c) Material used d) All of these	5	2
6	Value analysis is a _________________process. a) Remedial b) Preventive c) Continuous d) None of these	5	1

Questions carrying 5 marks

Sl No	Questions	CO	BTL
1	State the types of values.	5	2
2	What are the objectives of value analysis?	5	1
3	Define the meaning of value analysis.	5	1
4	Explain the process of value analysis.	5	1
5	Discuss Kaizen 5S model in an industrial concept.	5	1
6	Define cost control in a production context.	5	2
7	Discuss the importance and methods of cost control.	5	1
8	Explain the DARSIRI method of value analysis.	5	1
9	Discuss MUDA (waste) management.	5	2